

Deep Data Analysis

1. Unique Words by Rating

We collected all unique words used in reviews for each rating. Each rating level (1–5) has thousands of distinct words.

Insight: Customers use a very wide vocabulary to express their experiences, showing that opinions vary a lot between ratings.

```
# Function to clean and tokenize text
```

```
def tokenize(text):
    text = str(text).lower()          # lowercase
    text = re.sub(r'^a-z\s', '', text) # keep only letters & spaces
    words = text.split()
    return set(words)                 # return unique words in this review
```

```
# Dictionary to store unique words per rating
```

```
rating_words = {}
```

```
for rating in df['Rating'].unique():
```

```
    # Filter reviews for this rating
```

```
    reviews = df[df['Rating'] == rating]['Review']
```

```
    # Collect all unique words across reviews of this rating
```

```
    unique_words = set()
```

```
    for review in reviews:
```

```
        unique_words.update(tokenize(review))
```

```
    rating_words[rating] = unique_words
```

```
# Print results
```

```
for rating, words in rating_words.items():
```

```
    print(f"\nRating {rating} has {len(words)} unique words:")
```

```
    print(list(words)[:30]) # show first 30 words as sample
```

Rating 3.0 has 47900 unique words:

```
['kindergarten', 'takin', 'thanks', 'xxlit', 'bracelets', 'moreoften', 'satisfactory', 'customize', 'airiness',
'differentnew', 'twotwine', 'chance', 'curly', 'lazarus', 'aestheticsand', 'sonthe', 'hurrydid', 'poder',
'acceptablethe', 'incrediblebr', 'dvn', 'voluptuous', 'plastic', 'diffcuit', 'pumpkin', 'poconos',
'microphone', 'snugg', 'contamination', 'lackingbr']
```

Rating 5.0 has 42971 unique words:

```
['kindergarten', 'thanks', 'bracelets', 'xxlit', 'herb', 'satisfactory', 'wellloved', 'customize', 'mkbr', 'ddnt', 'statebr', 'attcl', 'revealingbr', 'curly', 'chance', 'altas', 'incrediblebr', 'poder', 'voluptuous', 'plastic', 'pumpkin', 'evitar', 'poconos', 'bluestriped', 'snugg', 'contamination', 'recognizes', 'pq', 'freely', 'coud']
```

Rating 4.0 has 48071 unique words:

```
['kindergarten', 'bibb', 'thanks', 'bracelets', 'weating', 'herb', 'satisfactory', 'waistmidsection', 'customize', 'ddnt', 'statebr', 'matronlywaistband', 'gstdadr', 'syntetic', 'briefcaselike', 'pillthin', 'lowlightdarkness', 'revealingbr', 'chance', 'curly', 'hipsy', 'altas', 'centernice', 'incrediblebr', 'poder', 'nanosuction', 'skirtcant', 'lalunesse', 'voluptuous', 'plastic']
```

Rating 2.0 has 41247 unique words:

```
['kindergarten', 'colorchallenged', 'takin', 'thanks', 'bracelets', 'problemey', 'attentiondrawing', 'satisfactory', 'customize', 'ddnt', 'attcl', 'jacke', 'revealingbr', 'chance', 'curly', 'neededweeks', 'beaming', 'inalei', 'goodbutthe', 'warthbr', 'poder', 'purposeoh', 'eventuallyi', 'voluptuous', 'lbls', 'plastic', 'pumpkin', 'whatsoeveri', 'unfourtantly', 'microphone']
```

Rating 1.0 has 42312 unique words:

```
['kindergarten', 'takin', 'thanks', 'xxlit', 'bracelets', 'needinterest', 'satisfactory', 'cuffpoor', 'customize', 'statebr', 'chlk', 'largemore', 'cox', 'chance', 'curly', 'washingsno', 'poder', 'microthin', 'voluptuous', 'plastic', 'pumpkin', 'evitar', 'microphone', 'timo', 'contamination', 'coud', 'freely', 'recognizes', 'pq', 'openthis']
```

2.Unique & Exclusive Words by Rating

Exclusive words are words that appear only in one rating and not in others.

Step 1: Collect all words per rating

```
rating_words = {}
```

```
for rating in df['Rating'].unique():
```

```
    reviews = df[df['Rating'] == rating]['Review']
```

```
    unique_words = set()
```

```
    for review in reviews:
```

```
        unique_words.update(tokenize(review))
```

```
    rating_words[rating] = unique_words
```

Step 2: Find words that are unique ONLY to each rating

```
exclusive_words = {}
```

```
all_ratings = set(rating_words.keys())
```

```
for rating in all_ratings:
```

```
    # words in this rating
```

```
    words_in_rating = rating_words[rating]
```

```
    # words in all other ratings
```

```
    words_in_others = set().union(*[rating_words[r] for r in all_ratings if r != rating])
```

```

# exclusive words = present only in this rating
exclusive_words[rating] = words_in_rating - words_in_others
# Results
for rating in rating_words:
    print(f"\nRating {rating}:")
    print(f" Total unique words = {len(rating_words[rating])}")
    print(f" Exclusive words   = {len(exclusive_words[rating])}")
    print(f" Sample exclusive words: {list(exclusive_words[rating][:30])}")

```

Rating 3.0:

Total unique words = 47900

Exclusive words = 17689

Sample exclusive words: ['againhe', 'cumulus', 'moreoften', 'fleecelines', 'whitr', 'gfshe', 'primed', 'cozy', 'cwaist', 'againwill', 'topcalf', 'airiness', 'itwrinkles', 'differentnew', 'currenciesbr', 'positionloosen', 'twotwine', 'anth', 'buyingnot', 'romperjumsuitone', 'justcant', 'saybetter', 'bigkept', 'inwear', 'lazarus', 'expectingmore', 'duas', 'customer', 'aestheticsand', 'sonthe']

Rating 5.0:

Total unique words = 42971

Exclusive words = 15583

Sample exclusive words: ['wellloved', 'lacingtying', 'gardeni', 'planetsgalaxies', 'mkbr', 'bigmakes', 'lcozy', 'inasmuch', 'buat', 'changedaccidentally', 'breezybr', 'easterwood', 'shoulf', 'waddup', 'epoch', 'mariner', 'designedi', 'parrish', 'bindlol', 'camsole', 'durableall', 'looserrelaxed', 'feetshort', 'birkenstockish', 'captor', 'pixelbypixel', 'troupe', 'doesso', 'belated', 'weadeatertrimmer']

Rating 4.0:

Total unique words = 48071

Exclusive words = 17919

Sample exclusive words: ['bibb', 'weating', 'preemptively', 'lfell', 'waistmidsection', 'directedbr', 'matronlywaistband', 'airportthankfully', 'utilitytactical', 'washedsmaller', 'glossshine', 'gstsdadr', 'syntetic', 'briefcaselike', 'complimentsquick', 'pillthin', 'lowlightdarkness', 'polyacrylic', 'colcl', 'hipsy', 'offjuzt', 'dbjskd', 'centernice', 'shimsaw', 'sssuuuppperr', 'nonjacking', 'upkimono', 'xlxxx', 'nanosuction', 'skirtcant']

Rating 2.0:

Total unique words = 41247

Exclusive words = 13605

Sample exclusive words: ['colorchallenged', 'paqer', 'problemy', 'cirles', 'loosefalling', 'attentiondrawing', 'messylooking', 'publicappropriate', 'tuile', 'backlash', 'braskin', 'flipflopbr', 'jacke', 'postexercise', 'morelonger', 'neededweeks', 'upay', 'precludes', 'sint', 'buckram', 'beaming', 'inalei', 'sweatstank', 'skinrub', 'insidefeels', 'goldand', 'goodbutthe', 'bondbr', 'warthbr', 'purposeoh']

Rating 1.0:

Total unique words = 42312

Exclusive words = 15255

Sample exclusive words: ['roo', 'needinterest', 'shineyglitter', 'ladrones', 'itlame', 'sugary', 'dirty muddy', 'cuffpoor', 'chlk', 'framw', 'overreacting', 'roundits', 'largemore', 'scanbr', 'purchasesent', 'matievil', 'cox', 'continuum', 'shrang', 'placecame', 'clothterrible', 'harmonic', 'sukana', 'washingsno', 'overeyeglasses', 'microthin', 'stayit', 'armillary', 'sisternothers', 'coolibars']

3.Sentiment Filtering of Exclusive Words

We filtered exclusive words into positive and negative groups. Rating 1 is dominated by negative exclusives like *'hurtful'* and *'shamelessly'*, while Rating 5 is dominated by positives like *'joyful'* and *'outstandingly'*.

Insight: Sentiment polarity strongly aligns with the rating value, confirming the validity of reviews.

Step 3: Sentiment filtering from exclusive words

```
import nltk
```

```
from nltk.corpus import opinion_lexicon
```

Download lexicon if not already installed

```
nltk.download('opinion_lexicon')
```

Positive and negative lexicon

```
positive_words = set(opinion_lexicon.positive())
```

```
negative_words = set(opinion_lexicon.negative())
```

```
exclusive_sentiment_words = {}
```

```
for rating in exclusive_words:
```

```
    pos_matches = exclusive_words[rating].intersection(positive_words)
```

```
    neg_matches = exclusive_words[rating].intersection(negative_words)
```

```
exclusive_sentiment_words[rating] = {
```

```
    "positive": list(pos_matches),
```

```
    "negative": list(neg_matches)
```

```
}
```

Results

for rating in exclusive_sentiment_words:

```
print(f"\nRating {rating} (exclusive sentiment words):")
print(f" Positive: {exclusive_sentiment_words[rating]['positive']}")
print(f" Negative: {exclusive_sentiment_words[rating]['negative']}")
```

Rating 1.0 (exclusive sentiment words):

Positive: ['dominates', 'subsidize', 'zenith', 'intelligible', 'enlighten', 'unrivaled', 'liberate', 'entice', 'rightfully', 'trustworthiness', 'unbound', 'auspicious', 'paramount', 'foolproof', 'relish', 'fascination', 'equitable', 'magnanimous', 'joyfully', 'levity', 'remission', 'tantalize', 'visionary', 'amenity', 'prominence']

Negative: ['hurtful', 'tawdry', 'shamelessly', 'refusal', 'braggart', 'dispiriting', 'misbehave', 'outrage', 'adamant', 'disagreement', 'inhumane', 'uncooperative', 'inflamm', 'delusion', 'laughingstock', 'accusing', 'farce', 'forged', 'feint', 'futility', 'insufferably', 'fruitlessly', 'madder', 'penalize', 'farfetched', 'scariest', 'quitter', 'abolish', 'brazen', 'unresolved', 'incite', 'fatally', 'consternation', 'acid', 'wayward', 'betrayal', 'poisonous', 'illegitimate', 'bungling', 'dirtbag', 'infringe', 'huckster', 'orphan', 'distracted', 'blasphemy', 'discriminate', 'helpless', 'implicate', 'inconsistence', 'exasperating', 'tortured', 'unkind', 'wreaks', 'insultingly', 'betrays', 'deplorable', 'sap', 'prejudice', 'dogged', 'breakup', 'distrusting', 'imbecile', 'dissatisfy', 'anomalous', 'snobby', 'misery', 'pathetically', 'incongruously', 'displease', 'averse', 'sneakily', 'shrill', 'hooligan', 'hostage', 'inadequately', 'fascist', 'endanger', 'evade', 'panick', 'despotic', 'bs', 'involuntarily', 'denies', 'worthlessly', 'improbable', 'ferociously', 'inextricably', 'mockery', 'liar', 'inconsequentially', 'demeaning', 'emphatically', 'worthlessness', 'incompetently', 'resigned', 'infernal', 'conceited', 'nonresponsive', 'contaminate', 'inconceivable', 'repugnant', 'repulsed', 'affront', 'ire', 'contravene', 'immovable', 'shortsighted', 'lethargic', 'distastefully', 'irredeemably', 'tyranny', 'haywire', 'condescending', 'stereotype', 'taxing', 'disparage', 'turbulent', 'racism', 'contempt', 'fucking', 'undercutting', 'disrespecting', 'sueing', 'undercut', 'callous', 'impending', 'predicament', 'raping', 'dissent', 'concede', 'starvation', 'offensively', 'insignificance', 'revulsion', 'boycott', 'dictator', 'unconfirmed', 'procrastinate', 'disillusioned', 'illegally', 'superficially', 'inescapable', 'disputed', 'inefficiency', 'idiocy', 'stumped', 'ridicule', 'listless', 'unimaginable', 'mortifying', 'impotent', 'cautionary', 'jumpy', 'vehemently', 'protested', 'clogged', 'unproven', 'fragmented', 'downturn', 'travesty', 'humiliating', 'lagging', 'unrelenting']

Rating 2.0 (exclusive sentiment words):

Positive: ['rectification', 'exhilaration', 'sweetly', 'indebted', 'gutsy', 'evaluative', 'fertile', 'ameliorate', 'fortitude', 'astonishingly', 'oasis', 'excellent', 'achievable', 'reclaim', 'dreamland', 'luckiest', 'renown', 'simplified', 'harmoniously', 'enrich', 'prestigious', 'elan', 'impressed', 'verifiable', 'restructuring', 'painlessly', 'continuity']

Negative: ['rampant', 'snare', 'harried', 'misaligns', 'disagreeable', 'misinterpret', 'barbaric', 'breach', 'unpopular', 'villain', 'kaput', 'oblivious', 'onerous', 'preposterous', 'mocked', 'starkly', 'incompetence', 'emphatic', 'nervousness', 'aghast', 'stubbornly', 'sloow', 'diametrically', 'bitchy', 'unwarranted', 'overblown', 'disarm', 'indeterminate', 'preposterously', 'slur', 'jaundiced', 'malignant', 'bewildered', 'contaminating', 'coerce', 'shamefully', 'suicidal', 'confound', 'meager', 'incapable', 'futile', 'tormented', 'contrived', 'chaotic', 'disarray', 'gibe', 'flighty', 'heartbreakingly', 'hysteric', 'anguish', 'inattentive']

'vibrated', 'nosey', 'uneventful', 'disservice', 'betray', 'grievous', 'incomparable', 'torturous', 'unsustainable', 'tamper', 'deadlock', 'fibber', 'spewed', 'inordinately', 'forgetfulness', 'slug', 'gawky', 'aversion', 'apocalyptic', 'uncaring', 'indecent', 'dearth', 'impatience', 'irate', 'omission', 'powerless', 'lambaste', 'inexorably', 'inundate', 'paltry', 'accusation', 'disorderly', 'perish', 'tingled', 'onslaught', 'unhappily', 'crippled', 'phobia', 'gullible', 'condemn', 'hatred', 'bereavement', 'dilapidated', 'outlaw', 'woeful', 'inflammatory', 'enflame']

Rating 3.0 (exclusive sentiment words):

Positive: ['earnest', 'dauntless', 'reasoned', 'flourishing', 'sensational', 'perseverance', 'supremacy', 'heartwarming', 'exuberance', 'dazzled', 'openly', 'judicious', 'effusive', 'outdo', 'privileged', 'blissfully', 'exuberant', 'mastery', 'bolster', 'gratifying', 'twinkly', 'ambitious', 'mightily', 'entrancing', 'richly', 'decisive', 'deference', 'refine', 'profuse', 'astonishment', 'fondness']

Negative: ['batty', 'falter', 'incessant', 'demonized', 'insinuate', 'irksome', 'tardy', 'calamity', 'declining', 'satirical', 'corrupt', 'criticized', 'traduce', 'degenerate', 'sluggish', 'ominous', 'indoctrination', 'runaway', 'interruption', 'delude', 'steeply', 'ruthless', 'overstates', 'rumple', 'skepticism', 'dissing', 'stew', 'exorbitantly', 'glower', 'uncontrolled', 'idle', 'antithetical', 'stodgy', 'madman', 'discouragement', 'complicit', 'regreted', 'misunderstand', 'murderer', 'taboo', 'gaffe', 'indignant', 'curt', 'grudge', 'unimaginably', 'mistrust', 'undependability', 'flagging', 'slog', 'sinfully', 'incongruous', 'dazed', 'deprived', 'aggrieved', 'stun', 'catastrophically', 'destruction', 'forbidding', 'unfulfilled', 'disasterous', 'unemployed', 'skulk', 'pittance', 'pedantic', 'brash', 'busybody', 'cannibalize', 'feckless', 'impoverished', 'vexing', 'lackadaisical', 'insinuating', 'mischievous', 'immaterial', 'misleadingly', 'derogatory', 'smugly', 'myth', 'unspeakable', 'confession', 'slaughtered', 'invader', 'altercation', 'misjudgment', 'rankle', 'caricature']

Rating 4.0 (exclusive sentiment words):

Positive: ['ardor', 'hotcake', 'truthfulness', 'amiable', 'zeal', 'witty', 'gratefully', 'advantageous', 'prudent', 'friendliness', 'panoramic', 'idolized', 'gloriously', 'noiseless', 'zest', 'overtaken', 'aspire', 'rectifying', 'undisputed', 'tenderly', 'agility', 'prosper', 'overtook', 'peppy', 'hardier', 'revel', 'admiration', 'empathy', 'outshone', 'humility', 'stimulating', 'divinely', 'mercifully', 'fresher', 'ovation', 'congratulate', 'unity', 'sweetness', 'hospitable', 'mesmerized', 'handily', 'luxuriant', 'rapture', 'hallowed', 'overtaking', 'playfully', 'enjoyably', 'upliftment', 'exalted', 'fanfare', 'convincingly', 'convience', 'sprightly']

Negative: ['overemphasize', 'contradict', 'barren', 'devilish', 'uncompromising', 'nastiness', 'resentment', 'admonishment', 'disadvantaged', 'inhibition', 'squirm', 'unjust', 'scandalized', 'mournful', 'balk', 'vicious', 'loath', 'adamantly', 'skittish', 'contradiction', 'funnily', 'leech', 'insufficiency', 'pervert', 'costlier', 'bonkers', 'underpowered', 'sulk', 'toil', 'inconsolable', 'despot', 'carnage', 'ambivalent', 'ineffectual', 'unskilled', 'commonplace', 'defiance', 'archaic', 'flounder', 'prohibitively', 'rebuke', 'viciously', 'devastation', 'rivalry', 'merciless', 'mischief', 'fateful', 'austere', 'bashful', 'quandary', 'entanglement', 'inexperience', 'obtuse', 'malaise', 'halfhearted', 'defy', 'frightful', 'intimidate', 'clouding', 'timidly', 'miscalculation', 'compulsion', 'gnawing', 'poverty', 'illicit', 'perverse', 'corrupted', 'subvert', 'debilitating', 'menacing', 'egregiously', 'tyrant', 'daze', 'bombastic', 'grumble', 'engulf', 'secretive', 'shameless', 'disown', 'uninformed', 'equivocal', 'contaminates', 'bereft', 'graft', 'stifle', 'plight', 'lecher', 'rabid', 'fearfully', 'misguide', 'fanatical', 'controversial', 'desolate', 'instability',

'arrogance', 'obsess', 'unexplained', 'douchbag', 'infirm', 'illogically', 'feign', 'tramp', 'obsessive', 'undetermined', 'decadence', 'scowl', 'depressingly', 'deluge', 'rape', 'untested', 'moan', 'unsound', 'demolish', 'bicker']

Rating 5.0 (exclusive sentiment words):

Positive: ['providence', 'relent', 'glimmering', 'elation', 'righteous', 'affirmation', 'smitten', 'fearless', 'stupendous', 'willingly', 'devout', 'politeness', 'exalt', 'outstandingly', 'faithfulness', 'responsibly', 'overture', 'terrifically', 'savior', 'angelic', 'upheld', 'covenant', 'outdone', 'ingeniously', 'booming', 'outperform', 'calmness', 'masterful', 'rejuvenated', 'simplifying', 'pinnacle', 'harmonious', 'harmonize', 'hale', 'eloquently', 'serene', 'fervent', 'enticingly', 'revolutionize', 'reverence', 'fruitful', 'empower', 'solace', 'coherent', 'handier', 'enthusiastically', 'rightful', 'convient', 'adroit', 'invincible', 'radiance', 'entertains', 'delectable', 'enviously', 'monumental', 'praising', 'pampers', 'commodious', 'meritorious', 'renowned', 'awed', 'joyful', 'empowerment', 'goood', 'momentous', 'courageous', 'surreal', 'fondly', 'majesty']

Negative: ['congested', 'overstate', 'frenetic', 'swagger', 'ail', 'monotony', 'kook', 'unsteadiness', 'oppressively', 'mobster', 'lurking', 'undermine', 'gloom', 'belated', 'douchebag', 'payback', 'arcane', 'irretrievable', 'submissive', 'dictatorial', 'unorthodox', 'caustic', 'overbearingly', 'cringes', 'doomsday', 'coward', 'suppress', 'frantically', 'grieving', 'uneasily', 'gawk', 'craftily', 'backwoods', 'protracted', 'stereotypically', 'surrender', 'covetous', 'hateful', 'hapless', 'impropriety', 'outsider', 'unmoved', 'addicting', 'criticizing', 'disdainfully', 'crabby', 'reproach', 'usurper', 'corrosive', 'commiserate', 'fainthearted', 'opponent', 'sidetracked', 'gruesome', 'insinuation', 'overshadow', 'wickedly', 'controversy', 'quack', 'enviously', 'beastly', 'terrorism', 'flareups', 'mania', 'protesting', 'lethal', 'startlingly', 'amputate', 'contrive', 'rift', 'zealot', 'stunt', 'compulsive', 'discredit', 'kooky', 'severity', 'eccentricity', 'lull', 'lackey']

4. Most Repeated Words by Rating

We checked the most frequently repeated words in reviews.

```
from collections import Counter
```

```
# Find most repeated words per rating
```

```
most_common_words = {}
```

```
for rating in df['Rating'].unique():
```

```
    reviews = df[df['Rating'] == rating]['Review']
```

```
    all_words = []
```

```
    for review in reviews:
```

```
        all_words.extend(tokenize(review))
```

```
        counter = Counter(all_words)
```

```
    most_common_words[rating] = counter.most_common(20) # top 20 words
```

```
# Show results
```

```
for rating, words in most_common_words.items():
```

```
    print(f"\nRating {rating}- Top repeated words:")
```

```
    for word, count in words:
```

```
print(f"{word}: {count}")
```

Rating 3.0- Top repeated words:

fit: 32041
like: 31046
size: 26838
small: 24746
look: 21272
would: 20441
wear: 19568
cute: 18589
nice: 17122
one: 16710
good: 16559
material: 16484
little: 15946
really: 14595
im: 13904
color: 13735
quality: 13282
ordered: 13146
great: 12709
get: 12215

Rating 5.0- Top repeated words:

love: 37554
fit: 32508
great: 31120
perfect: 17427
like: 16476
look: 16077
well: 15980
quality: 15763
good: 15398
size: 15185
wear: 15021
nice: 14960
comfortable: 14379
color: 13547
one: 11213
cute: 11168
bought: 11098
really: 10436
would: 10267
dress: 10229

Rating 4.0- Top repeated words:

fit: 36426
like: 29700

little: 25320
size: 24777
great: 24748
good: 23371
nice: 22856
love: 21276
look: 20131
wear: 19842
would: 19520
cute: 18288
small: 17334
well: 17092
really: 15738
im: 15006
one: 14920
color: 14825
bit: 14475
quality: 13873

Rating 2.0- Top repeated words:

like: 23805
small: 22774
fit: 19608
size: 19084
look: 15440
would: 14184
wear: 14093
one: 13807
material: 12974
way: 11090
ordered: 10928
im: 10015
really: 10009
quality: 9732
get: 9375
disappointed: 8873
color: 8772
even: 8711
back: 8338
dont: 8267

Rating 1.0- Top repeated words:

like: 23639
small: 20232
size: 16444
fit: 15825
one: 14589

look: 14108
would: 12993
even: 11926
material: 11764
quality: 11752
dont: 11086
disappointed: 11080
way: 11054
wear: 11050
ordered: 10993
cheap: 10323
made: 9856
money: 9551
get: 9319
time: 8943

5. Product Mentions in Reviews

By scanning for product keywords, we found which fashion products are most reviewed.

Product Review_Count

0	shirt	36841
2	top	34144
1	bag	20992
6	skirt	7027
4	jacket	6148
3	tshirt	5975
8	hoodie	2118
7	blouse	2022
9	glasses	376
5	shoes	290
11	shorts	236
14	leggings	152
10	jeans	133
12	sandals	41
16	saree	14
13	kurti	12
15	trousers	2

